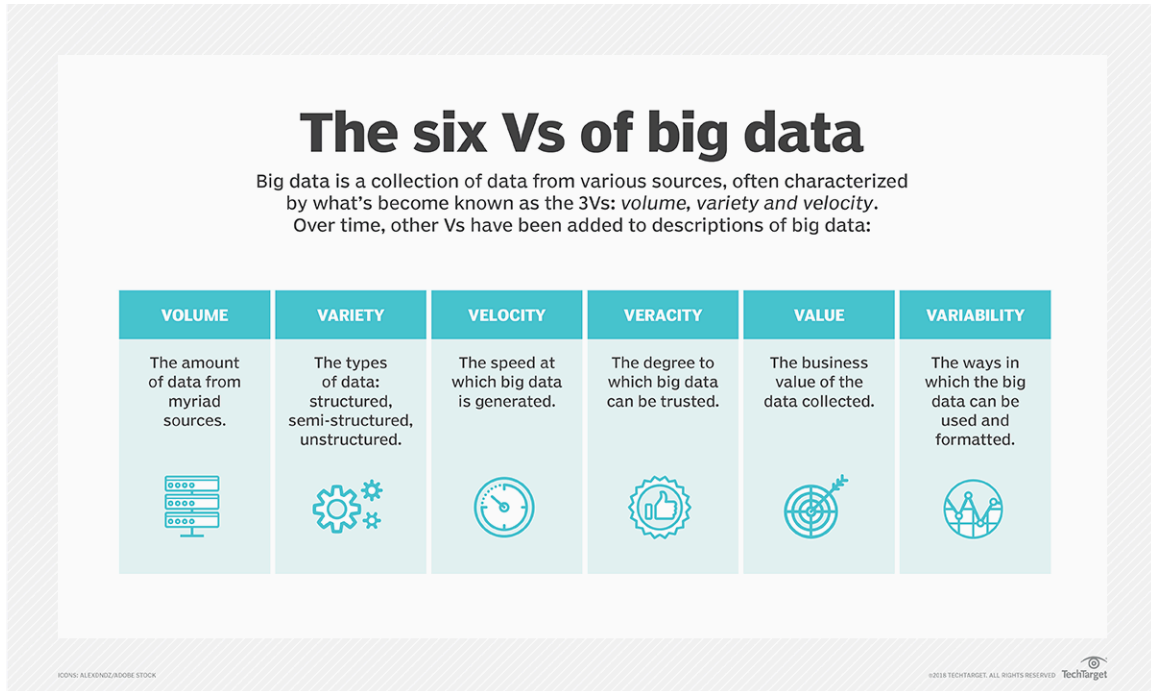


EL10: Biggish data

Big data



Biggish data

Our focus:

- Only tabular data
- Original data might fit into physical RAM but ...
- might be multiplied due to temporary necessity
- physical RAM might be constrained due to other processes/settings
- things might just take too much time ... and life is short ...

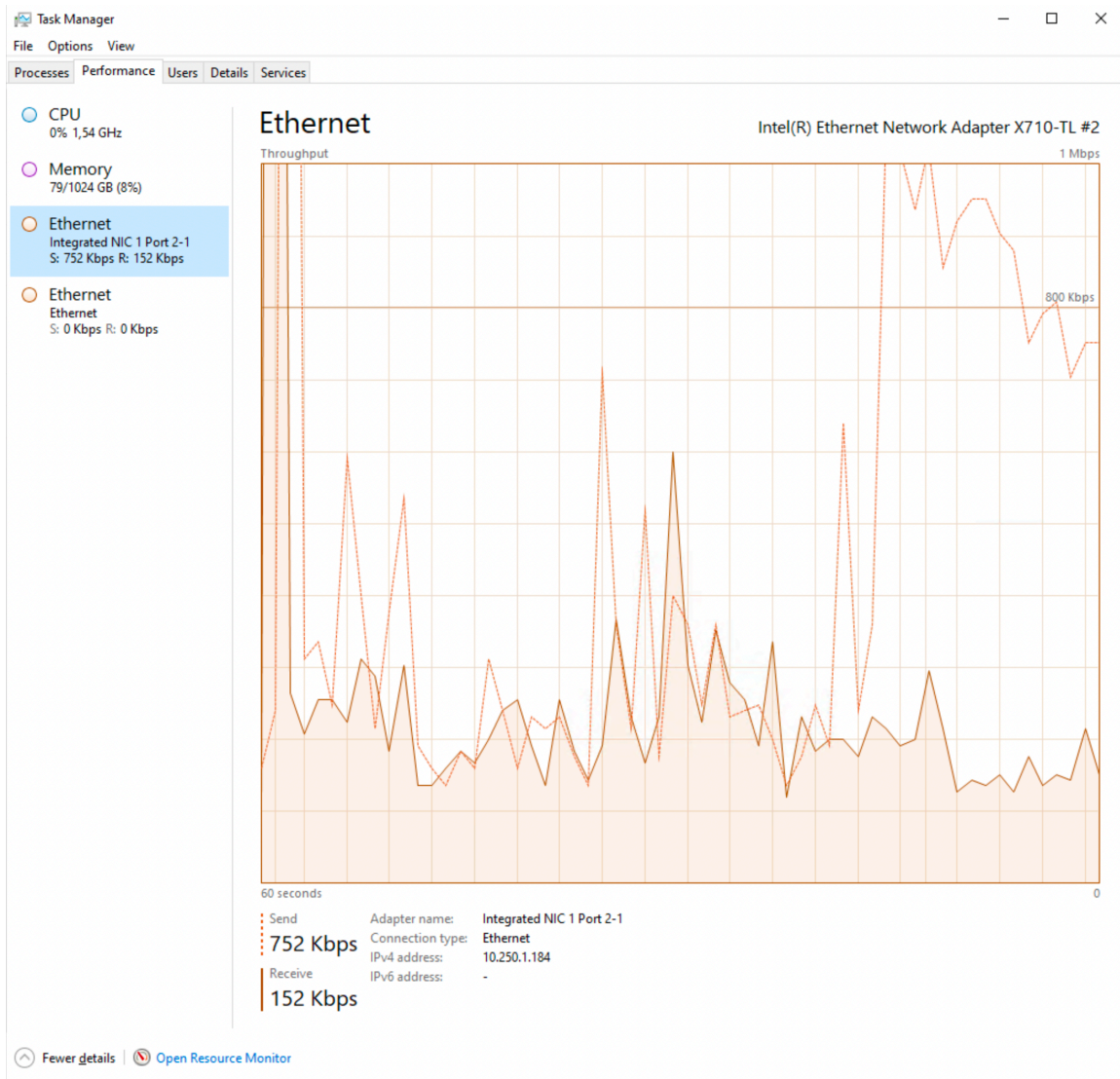
Software

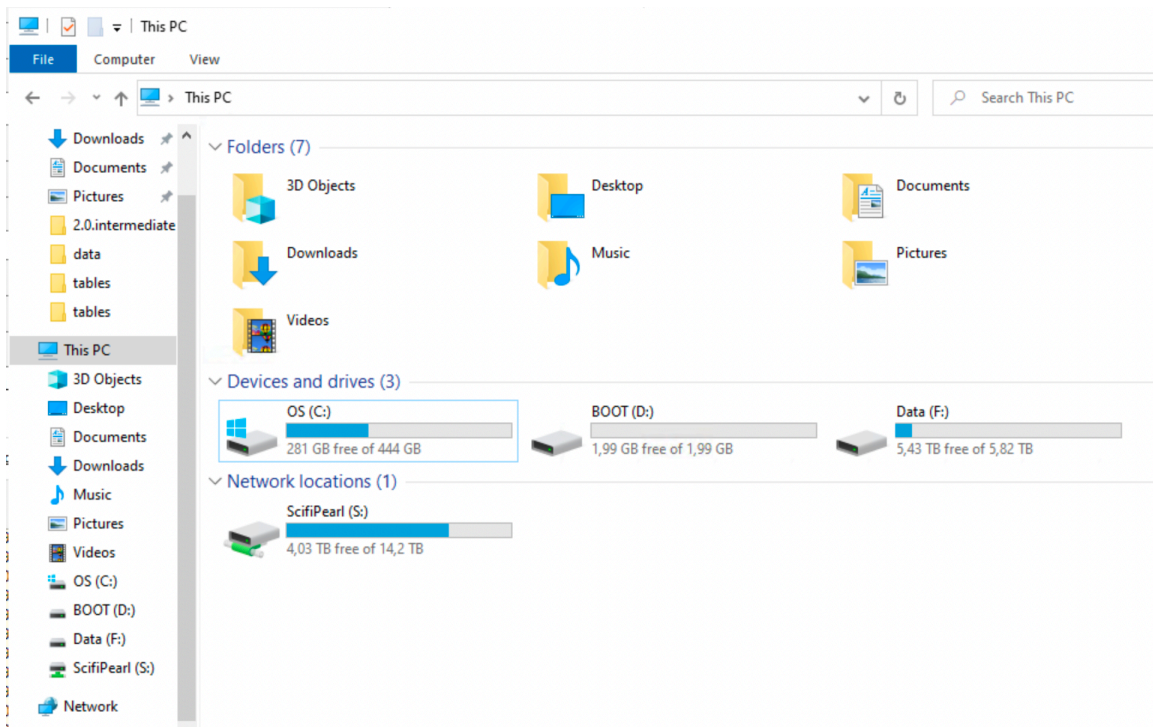
- SAS for big data files from register holders
- R for everything else
- Database?

File storage

- Avoid reading big files over ethernet connection
 - data stored on network drive
- Usually faster to first copy file to local SSD drive

- `fs::file_copy(path, new_path)`
 - (“*libuv provides a wide variety of cross-platform sync and async file system operations.*”)
 - obviously only if “local drive” is also in the (same) secure environment
 - work with the project on disk
 - move back
 - Although not always possible with very large files it seems
-

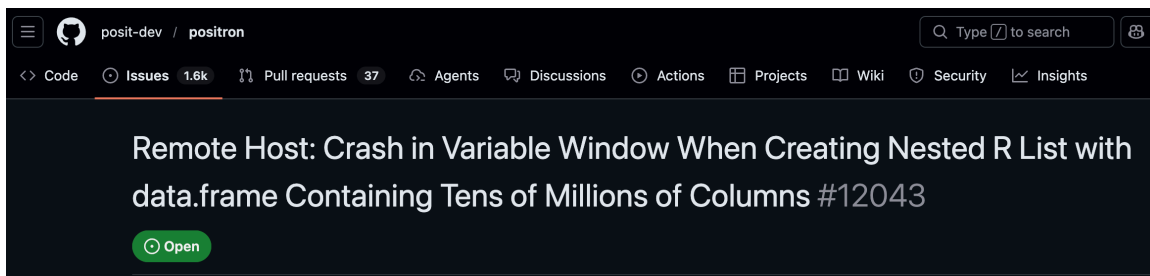




Comparison vs base equivalents

`fs` functions smooth over some of the idiosyncrasies of file handling with base R functions:

- Vectorization. All `fs` functions are vectorized, accepting multiple paths as input. Base functions are inconsistently vectorized.
- Predictable return values that always convey a path. All `fs` functions return a character vector of paths, a named integer or a logical vector, where the names give the paths. Base return values are more varied: they are often logical or contain error codes which require downstream processing.
- Explicit failure. If `fs` operations fail, they throw an error. Base functions tend to generate a warning and a system dependent error code. This makes it easy to miss a failure.
- UTF-8 all the things. `fs` functions always convert input paths to UTF-8 and return results as UTF-8. This gives you path encoding consistency across OSes. Base functions rely on the native system encoding.
- Naming convention. `fs` functions use a consistent naming convention. Because base R's functions were gradually added over time there are a number of different conventions used (e.g. `path.expand()` vs `normalizePath()`; `Sys.chmod()` vs `file.access()`).



[Still open?](#)

```
Rterm
C:\Windows\system32>"C:\Program Files\R\R-4.4.1\bin\R.exe"

R version 4.4.1 (2024-06-14 ucrt) -- "Race for Your Life"
Copyright (C) 2024 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

Loading required package: digest
Loading required package: tibble
> _
```

ALTREP

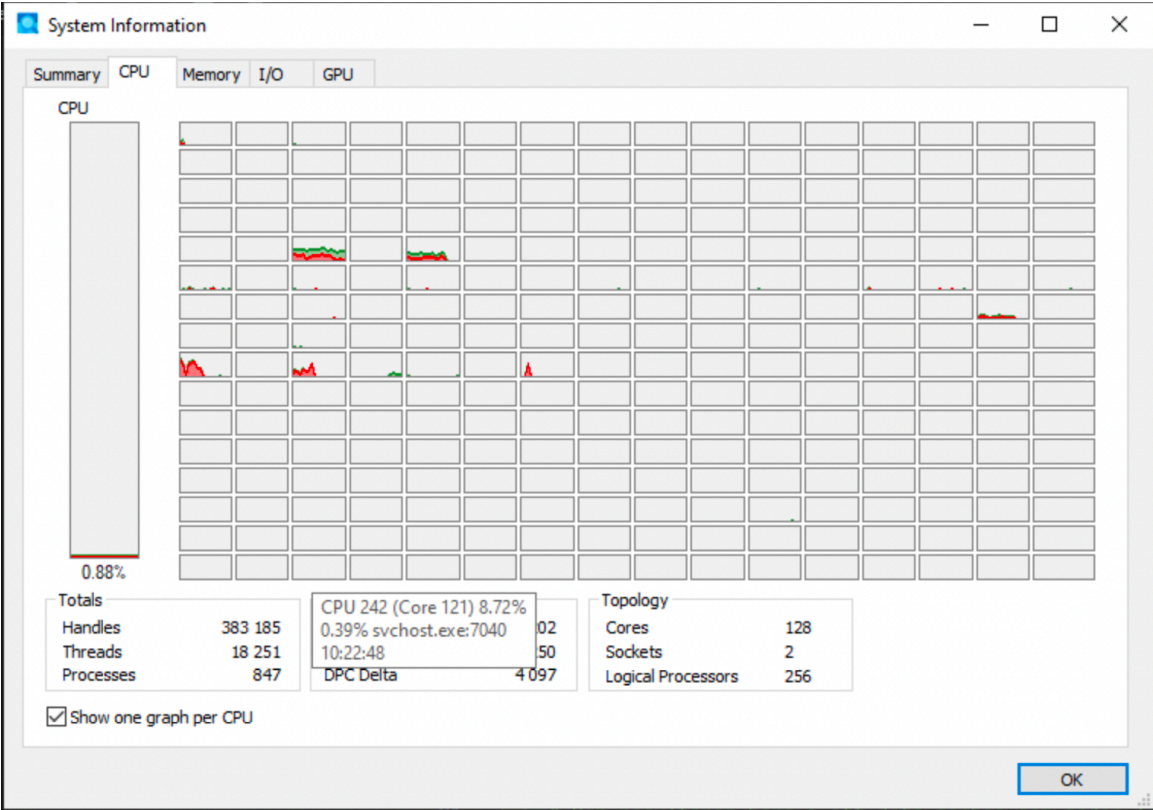
- **ALT**ernative **RE**presentation of R Objects
- Since R version 3.5.0
- Applies to the C-level representation of R objects under the hood
- Used by some base R packages/functions + specialized packages

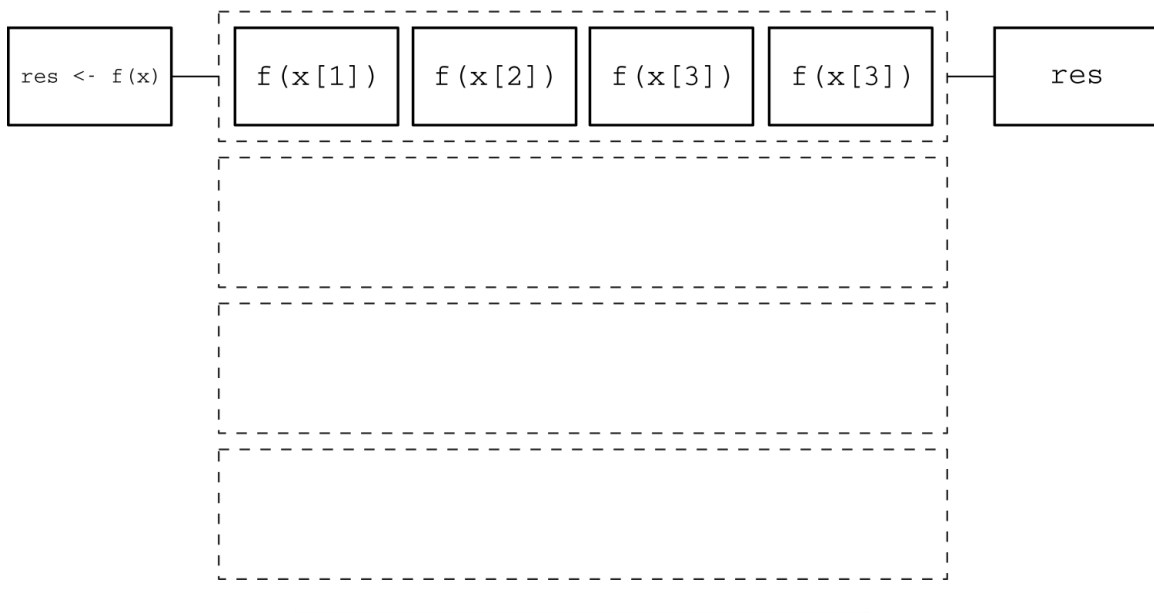
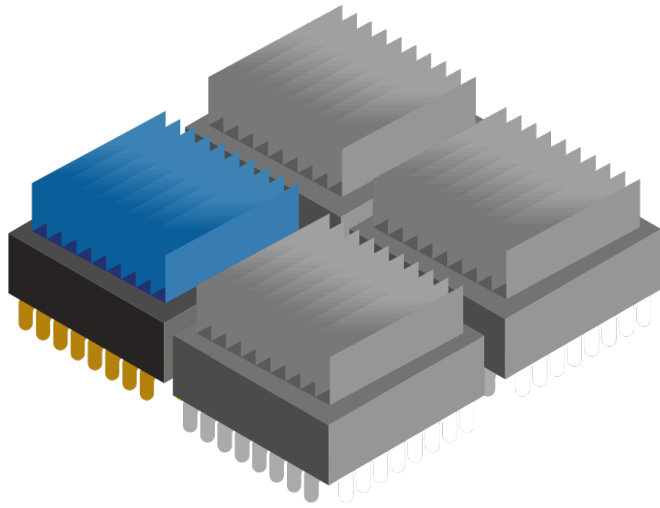
allow vector data to be in a memory-mapped file; distributed, e.g. within Apache Spark or Hadoop; shared with other applications, e.g. with Apache Arrow; allow compact representation of arithmetic sequences; allow adding meta-data to objects; allow computations/allocations to be deferred; support alternative representations of environments.

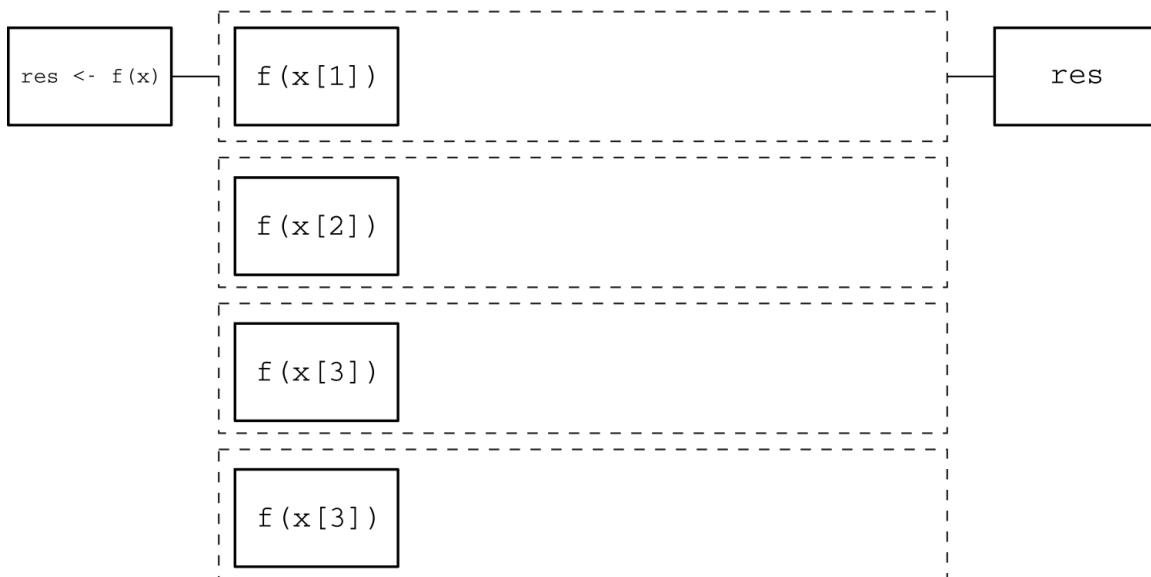
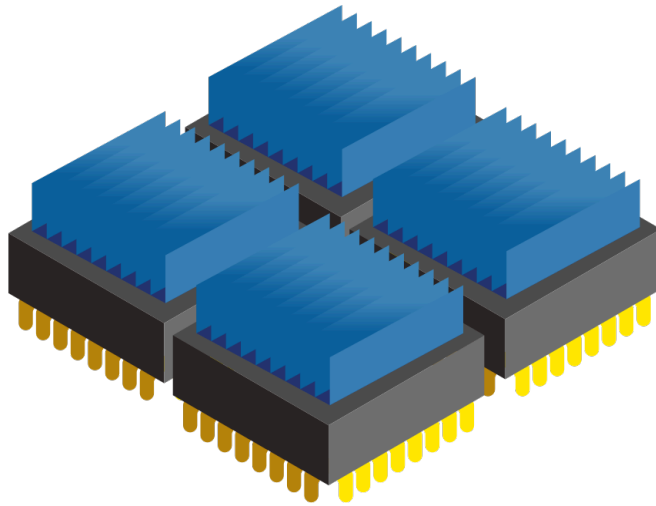
Dates

- Use `as.IDate()` instead of `as.Date()` if using `{data.table}`. Integer based.
- Use `{anytime}` (or maybe `{fasttime}`) if conversion is needed (ALTREP)

Paralellization







Parallelization

- R is single threaded by default
- Some packages and functions may use multithreading by default
- Some functions might have arguments such as `nthreads`, `ncores` etc
- To handle multicore/thread computations can be (very) difficult (I've heard)
- Easy if data can be divided and conquered (split-apply-combine)
- `{futureverse}` is my favorite (Henrik Bengtsson)
 - `{furrr}` is a drop-in replacement for `{purrr}`

`{furrr}`

```
# How many cores do we have?  
parallel::detectCores()
```



```
[1] 12
```

```
library(furrr)
```

Loading required package: future

```
# Save at least one for OS etc
plan(multisession, workers = parallel::detectCores() - 1)

1:10 |>
  future_map(rnorm, n = 10, .options = furrr_options(seed = 123)) |>
  future_map_dbl(mean)
```

```
[1] 1.180279 2.140442 2.909823 3.692207 5.058100 6.653926 7.065630 7.960713
[9] 9.105674 9.766827
```

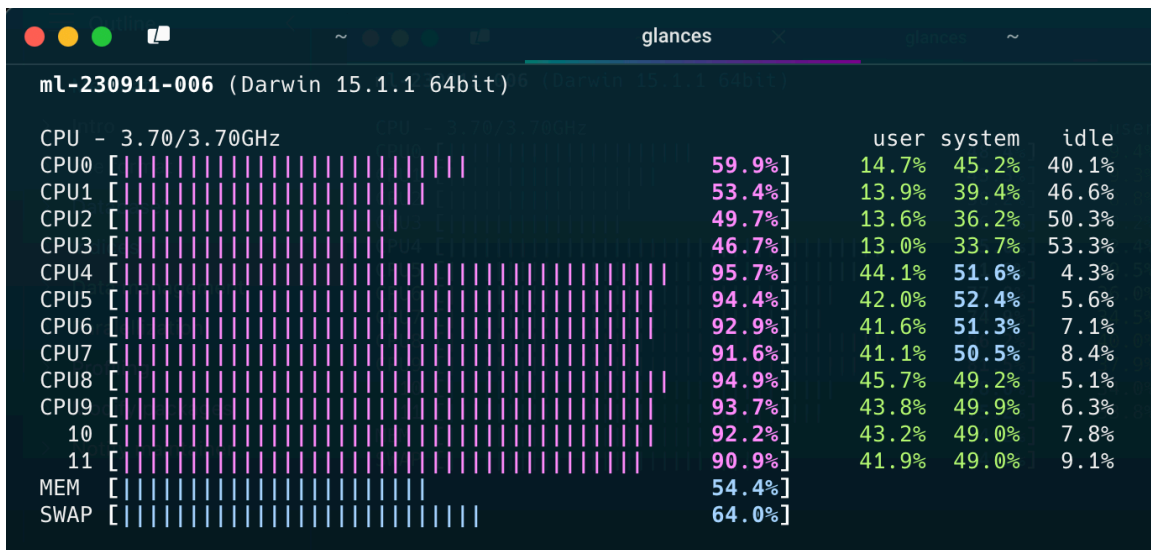
{mirai}

- Suposed to be up to 1,000 times “faster” compared to {furrr} (according to some metric ...)
- should not block the main process while executing (haven’t tried)

Designed for simplicity, a ‘mirai’ evaluates an R expression asynchronously in a parallel process, locally or distributed over the network, with the result automatically available upon completion.

```
library(mirai)
library(data.table)
daemons(11)

# data.table with 1 billion rows:
dt <- data.table(x = seq_len(1e9), y = rnorm(1e9))
# row sums (row wise )
mirai_map(dt, sum)[.flat]
```



{purrr}

- Introduced in `in_parallel()` in 2025
- Based on `mirai` but without the specialised syntax
- Needs to be explicit on which objects/functions to export to each worker/daemon

```
library(purrr)
library(mirai)

# Set up parallel processing (6 background processes)
daemons(6)

# Sequential version
mtcars |> map_dbl(\(x) mean(x))
```

mpg	cyl	disp	hp	drat	wt	qsec
20.090625	6.187500	230.721875	146.687500	3.596563	3.217250	17.848750
vs	am	gear	carb			
0.437500	0.406250	3.687500	2.812500			

```
#>   mpg   cyl  disp    hp  drat    wt   qsec    vs    am   gear
carb
#> 20.09   6.19 230.72 146.69   3.60   3.22  17.85   0.44   0.41   3.69
2.81
```

```
# Parallel version - just wrap your function with in_parallel()
mtcars |> map_dbl(in_parallel(\(x) mean(x)))
```

```

      mpg      cyl      disp      hp      drat      wt      qsec
20.090625  6.187500 230.721875 146.687500  3.596563  3.217250 17.848750
      vs      am      gear      carb
0.437500  0.406250  3.687500  2.812500

#>      mpg      cyl      disp      hp      drat      wt      qsec      vs      am      gear
carb
#> 20.09  6.19 230.72 146.69  3.60  3.22 17.85  0.44  0.41  3.69
2.81

# Don't forget to clean up when done
daemons(0)

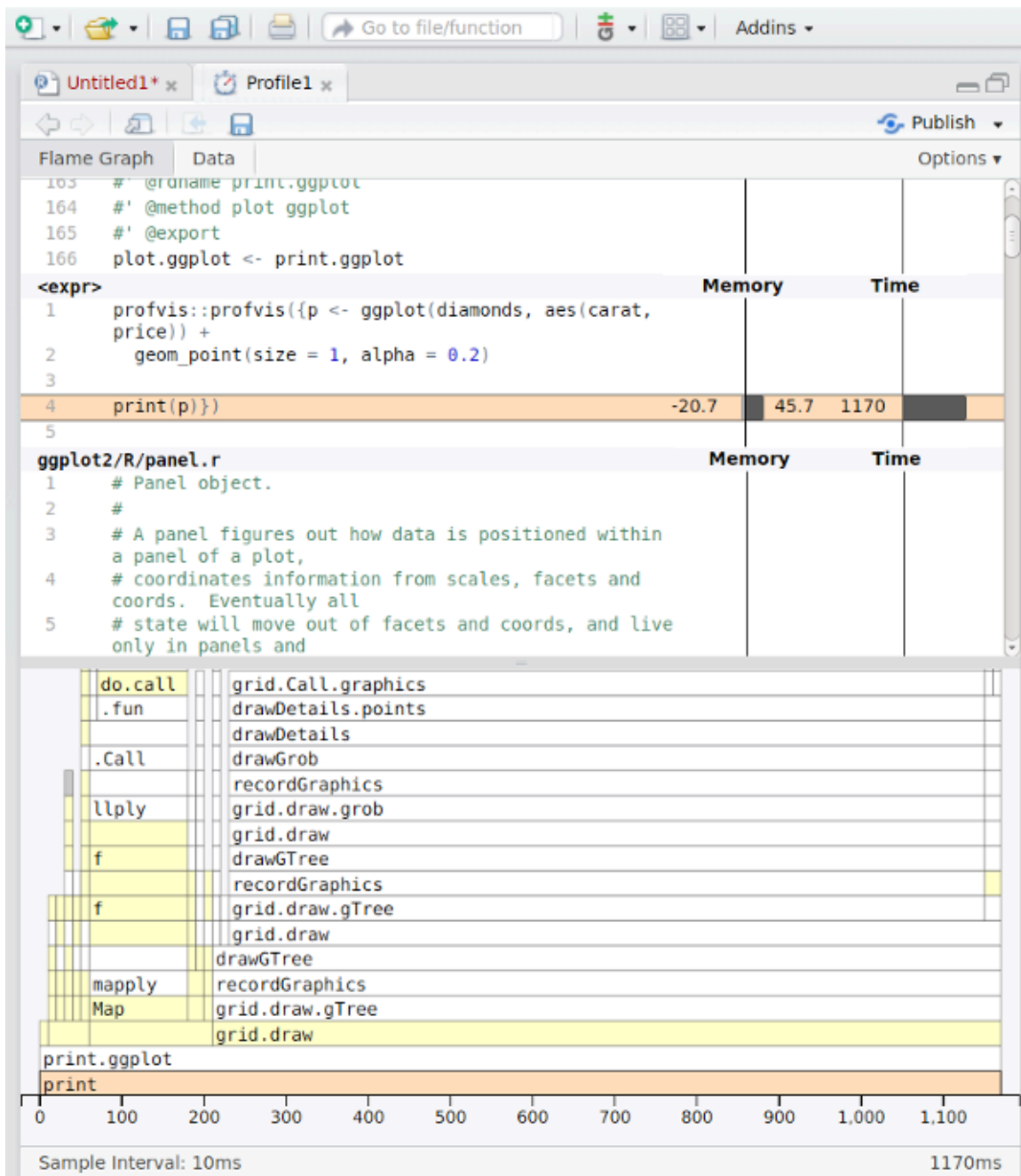
```

Note

- Data must be copied for each parallel worker
- If all data is needed for each worker, it will be multiplied in memory
 - This might not be possible for big data
 - It also takes time
- Hence, there is no guarantee that the over all time will decrease
- Often a tradeoff between no of workers/threads/deamons/cores and efficiency etc
- “Standard” computations in {data.table} is optimized for single core
- Message handling and progress bar etc is complicated

Profiling

- First version of R script might be inefficient
 - Optimazation is difficult
 - Might be suboptimal if made ad hoc
 - {profvis} visualise time and memory usage for each step
 - improve the most important step first
 - For example change {base} and {tidyverse} code to {data.table}
 - set keys
 - Efficient handling of dates and strings
 - itterate
 - Integrated in RStudio (but not yet Positron)
-



Modify packages

Modify packages

- Functions from packages might be inefficient
- Profile those as well
- Clone package from GitHub if available or download source code from CRAN
- Improve

- Just load the modified function in global environment
 - might need to modify internal calls to `:::-` accessed functions
- Or rebuild/install the package if more convenient

! Notify maintainer

If you find ways to improve a package, the maintainer might be eager to hear! Suggest pull request or open GitHub issue!

C-code changes

- If the package use C-code it is probably already very fast
- If not, and you can fix it, the package must be re-compiled
- OS dependent
- Might be tricky if using restricted environments (TRE?)
- Can still be done by rhub (might need to change maintainer-e-mail temporarily)

rhub 2.0.0 Reference Articles News

rhub

R-hub v2

R-hub v2 uses GitHub Actions to run `R CMD check` and similar package checks. The `rhub` package helps you set up R-hub v2 for your R package, and start running checks.

- Installation
- Usage
 - Requirements
 - Private repositories
 - Setup
 - Run checks
- The R Consortium runners
 - Limitations of the R Consortium runners
- Code of Conduct
- License

Installation

Install `rhub` from CRAN:

```
pak::pkg_install("rhub")
```

Links

- [View on CRAN](#)
- [Browse source code](#)
- [Report a bug](#)

License

[MIT + file LICENSE](#)

Community

[Code of conduct](#)

Citation

[Citing rhub](#)

Developers

Gábor Csárdi
Author, maintainer

Maëlle Salmon
Author

Funder

fastverse 0.3.4
DOCUMENTATION
VIGNETTES
NEWS
R-UNIVERSE
BLOG

Search for

fastverse

The *fastverse* is a suite of complementary high-performance packages for statistical computing and data manipulation in R. Developed independently by various people, *fastverse* packages jointly contribute to the objectives of:

- Speeding up R through heavy use of compiled code (C, C++, Fortran)
- Enabling more complex statistical and data manipulation operations in R
- Reducing the number of dependencies required for advanced computing in R

The **fastverse** package is a meta-package providing utilities for easy installation, loading and management of these packages. It is an extensible framework that allows users to (permanently) add or remove packages to create a 'verse' of packages suiting their general needs, or even create separate 'verses' of their own.

fastverse packages are jointly attached with `library(fastverse)`, and several functions starting with `fastverse_` help manage dependencies, detect namespace conflicts, add/remove packages from the *fastverse* and update packages. The [vignette](#) provides a concise overview of the package.

Links

[View on CRAN](#)
[Browse source code](#)
[Report a bug](#)

License

[GPL-3](#)

Citation

[Citing fastverse](#)

Developers

Sebastian Krantz
Author, maintainer
[More about authors...](#)

Bibliography